Implementation of Shearing via an Adaptive Pressure-Feedback Loop for the Simulation and Continuum Analysis of Granular Flow

Poster ID: 95 | Course: ME299 | Siddhantkumar Thute | Advisor: Harkirat Singh | Jan-Apr 26

Problem Statement

Granular materials like powders and grains are industrial staples, but their flow behavior is complex. While Discrete Element Method (DEM) simulations (via LAMMPS) provide high physical accuracy, they are computationally expensive and produce "noisy" discrete data that is difficult to use for engineering design. The Project Goal is to bridge the gap between microscopic grain interactions and macroscopic flow analysis by developing a high-fidelity simulation and post-processing pipeline.

- Advanced Simulation: Implemented a Triclinic box for infinite shear and a custom Pressure Feedback Loop to maintain a constant normal stress (P = 100 Pa).
- The Data Challenge: Raw simulation outputs consist of individual particle "dots." These must be converted into smooth, continuous fields to be useful for continuum mechanics.
- Solution: We generated a robust dataset through multiple flow simulations and applied pre-defined post-processing structures to extract smooth continuum profiles from noisy data.

This processed dataset is used to train and validate a Neural Network for real-time flow prediction (Refer to Poster No. 93).

METHODOLOGY

1. Simulation Physics & Shear: We initiated the study using compressed restart files from the mid-semester "pouring" phase. To study flow rheology, a Triclinic box was utilized, allowing for continuous horizontal shear (erate) across a range of 0.1 s^{-1} to 5.0 s^{-1} (under 0 gravity).
While the initial phases used fixed boundaries for containment, the shearing stage implemented periodic X-boundaries to simulate an infinitely long flow and achieve steady-state rheology. This transition removed edge effects, allowing for the characterization of "bulk" material behavior.

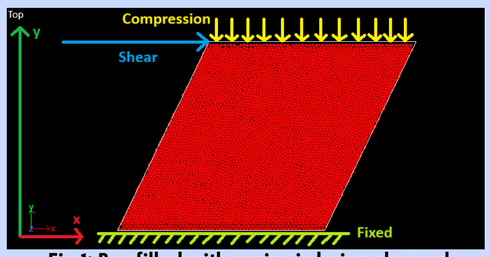

Fig 1: Box filled with grains is being sheared

Compute Instantaneous Pressure (P_{inst}) → Pressure Error Analysis (P_{target} - P_{inst}) → Apply Dual-Axis Deformation (fix deform).

2. Dynamic Stability (Pressure Feedback): A static box volume during shear often leads to unrealistic pressure spikes or grain "explosions." To mimic real-world conditions, a custom feedback loop was integrated:

- The system constantly compares instantaneous pressure to a 100 Pa target.
- The top wall height is dynamically adjusted using a clamped displacement algorithm to maintain contact stability and constant density.

3. Continuum Mapping (Post-Processing): Raw particle data (discrete) was converted into continuous engineering fields using an established MATLAB framework. By applying Lucy and Gaussian spatial kernels across 600 vertical slices, we extracted smooth macroscopic profiles for velocity, shear rate, and normal stress, removing the numerical noise of individual grain interactions.

Steady-State Data Dump (Triggered after 300k steps for equilibrium). → MATLAB Post-Processing Logic. → Final Output (profiles for Velocity, velocity gradient and Stress).

WORKFLOW

Input Setup (Triclinic Geometry) → Physics Simulation (Hookean Contact + Tangential Friction) → Control Loop (Real-time 100 Pa Pressure Feedback) → Data Extraction (Steady-State Dumps post-300k steps) → Post processing data (Used standard functions used to handle noisy data) → Continuous Field Output (Velocity and Stress profiles)

RESULTS - Post processing

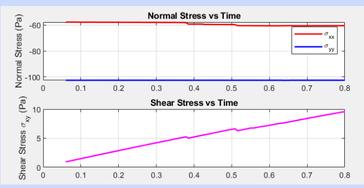

Fig 2: Average Stress vs time for shear rate = 0.1 s^{-1}

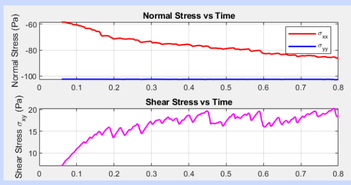

Fig 3: Average Stress vs time for shear rate = 1 s^{-1}

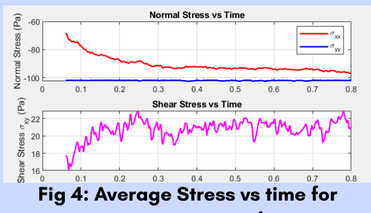

Fig 4: Average Stress vs time for shear rate = 5 s^{-1}

The confining normal stress σyy is maintained at -100 Pa throughout all simulations by the pressure-feedback wall, confirming effective pressure control. The lateral stress σxx starts near -60 Pa at low shear rates but progressively decreases (becomes more compressive, reaching ˜-90 Pa) as shear rate increases. The stress ratio σxx/σyy ≠ 1 at all rates, revealing that the granular assembly is not isotropic under shear.

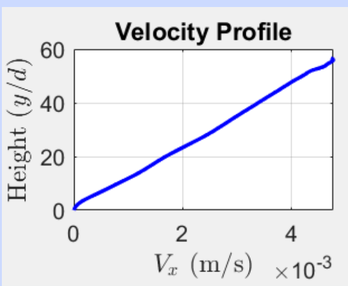

Fig 5: Velocity Profile for shear rate = 0.1 s^{-1}

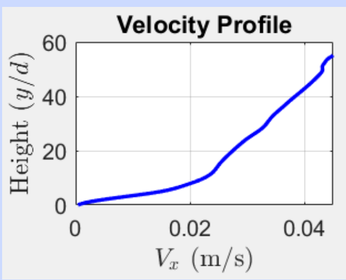

Fig 6: Velocity Profile for shear rate = 1 s^{-1}

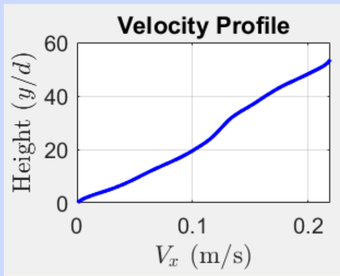

Fig 7: Velocity Profile for shear rate = 5 s^{-1}

At $\dot{\gamma}$ = 0.1 s^{-1}, the velocity profile Vx(y) is linear, confirming homogeneous simple shear with a uniform shear rate across the full height. As shear rate increases to 0.5 s^{-1}, the profile develops a gentle S-shape, with slower grains near the base and faster grains concentrated near the top (observed from the simulation video as well as the above graphs).

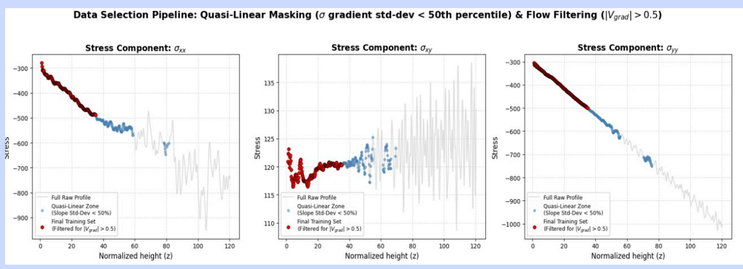

Fig 8: All 3 components of stresses vs height for the training data

- These profiles in Fig 8 show vertical stress gradients caused by gravity in the training set, differing from the gravity-free LAMMPS environment.
- Despite the gravity difference, both datasets share fundamental compression mechanics, allowing the model to generalize normal stress predictions effectively .

RESULTS - Neural network prediction

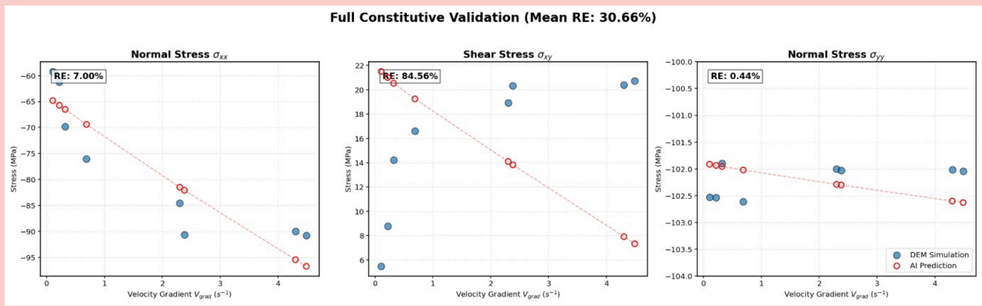

Fig 9: Validation of Neural Network Predictions against the DEM Data

1. High-Precision Prediction (Normal Stresses)

- Performance: Excellent accuracy in predicting σ_{yy} (RE: 0.44%) and σ_{xx} (RE: 7.0%).
- Significance: This confirms the model successfully generalized the relationship between velocity gradients ($v_{x,grad}$) and internal pressure.
- Link to Simulation: The near-perfect match in σ_{yy} validates the Pressure Feedback Loop implemented in the LAMMPS code, proving that the consistent physical environment allowed the AI to predict the flow state accurately.

2. Explaining Shear Stress Deviation (σ_{xy})

- Observation: A higher error (RE: 84.6%) exists in the absolute magnitude of shear stress, though the qualitative trend (slope direction) remains correct.
- Reason: The 84.56% shear error stems from model bias: the NN assumes gravity-linked, depth-dependent pressure that is absent in our boundary-controlled, zero-gravity environment. Lacking explicit pressure inputs, the model identifies qualitative trends through velocity gradients but fails to predict absolute magnitudes. Normal stress remains accurate (<10%) because its relationship with local inter-particle confinement remains physically robust across both flow regimes.

CONCLUSION & FUTURE SCOPE

We have established a high-fidelity pipeline for granular flow by integrating a custom pressure feedback loop within a Triclinic LAMMPS environment. This ensured steady-state stability (100 Pa) across shear rates from 0.1 s^{-1} to 5.0 s^{-1}, effectively preventing numerical instabilities and grain "explosions." The MATLAB framework successfully bridged the discrete-to-continuous gap using Lucy and Gaussian kernels.
In future, there is significant scope to extend this to 3D volumetric assemblies and non-spherical grain geometries and generate data for those flows. Further develop and optimize neural network architectures to achieve high-fidelity predictions across all rheological parameters, with targeted improvements in modeling friction-dependent shear stress.

REFERENCES

- LAMMPS Documentation
- Lucy, L. B. (1977). "A numerical approach to the testing of the fission hypothesis." The Astronomical Journal, 82, 1013-1024.